Model Research and Selection

AI Response Quality Evaluator Agent — Multi-Agent Hallucination Detection & Validation System

Milestone 1 Deliverable
Architecture & LLM Selection

1 Introduction & Project Context

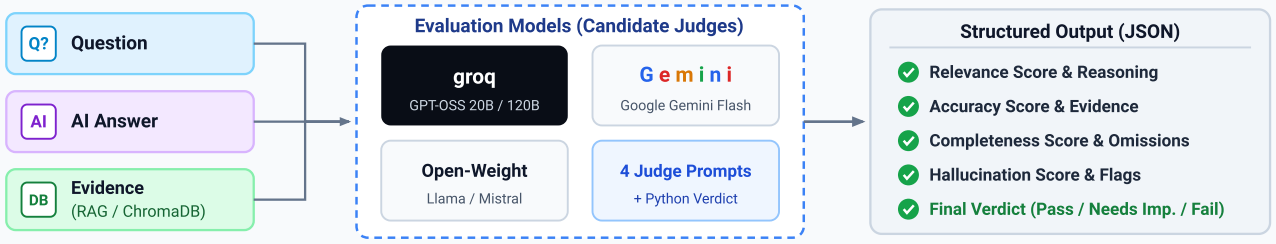
The **AI Response Quality Evaluator Agent** requires a language model that can evaluate AI-generated answers for relevance, factual accuracy, completeness and hallucinations. The model must also compare answers against evidence retrieved through the RAG pipeline and return structured evaluation results.

As AI-powered applications become increasingly common, manual evaluation is slow, inconsistent, and impossible to scale. The following models are considered for this project based on their capabilities, cost, speed and suitability for evidence-based answer evaluation across single-response and automated CSV batch evaluation modes.

5-AGENT JUDGING PIPELINE ARCHITECTURE

- **Relevance Judge Agent:** Scores how directly the answer addresses the question.
- **Accuracy Judge Agent:** Checks factual correctness against reference chunks.
- **Hallucination Detection Agent:** Flags unsupported claims vs. RAG evidence.
- **Completeness Judge Agent:** Checks whether all question aspects are covered.
- **Verdict Agent:** Aggregates weighted scores into Pass / Needs Improvement / Fail.

Model Evaluation in the Project — End-to-End Multi-Agent RAG Flow



2 Models Considered

GPT-OSS 20B (Groq) — Fast & Cost Effective

A compact open-weight model available through the Groq API. Supports reasoning and structured JSON output for rapid single and batch evaluation.

GPT-OSS 120B (Groq) — Higher Cost

A larger model with stronger reasoning capabilities. Useful for complex evidence comparison and subtle multi-hop claim verification.

Google Gemini Flash — Hosted API

A hosted model designed for fast, general-purpose tasks. Suitable for relevance checking and structured response generation.

Llama / Mistral (Open-Weight) — Local / Self-hosted

Open-weight models that can be deployed locally or on self-hosted infrastructure for offline or custom hardware setups.

Original Visual Reference Architecture: Full preservation of the project's baseline model evaluation diagram, selection badges, and RAG verification pipeline.



3 Model Comparison & Project Suitability Matrix

Criteria	GPT-OSS 20B (Groq) ★	GPT-OSS 120B (Groq)	Gemini Flash (Google)	Llama / Mistral (Open-weight)
Speed	**Very high** (LPU accelerated)	High	Model-dependent	Hardware-dependent
API cost	**Lower** (Budget-friendly)	Higher	Verify current pricing	Hosting-dependent
Structured output	**Supported** (Strict JSON)	Supported	Verify selected model	Serving-dependent
Deployment	Groq API	Groq API	Google API	Local or hosted
Multi-Agent & CSV Batch Fit	**Ideal** (Handles 4 calls/row with minimal latency)	Slower & costlier on large CSV batches	Good throughput; watch rate limits on free tiers	Requires dedicated GPU server for batch concurrency
RAG Grounding (SQuAD / TruthfulQA)	**Strong** factual & hallucination flagging	Very strong on nuanced multi-hop claims	Strong general context comprehension	Varies by quantization & parameter size
Project suitability	**Initial prototype & primary pipeline**	Quality comparison	Alternative benchmark	Self-hosted option

4 Selection Criteria

Factual accuracy
Correctly identifies factual errors using retrieved evidence (Accuracy Judge Agent).

Hallucination detection
Identifies unsupported claims while minimizing false alarms (Hallucination Detection Agent).

Relevance and completeness
Checks whether answers address the question adequately without omission.

Consistency
Produces stable per-dimension scores, reasoning explanations, and valid structured JSON output.

Speed and cost
Supports efficient single-answer and automated CSV batch evaluation without high latency.

Integration
Works seamlessly with Python, ChromaDB-based retrieval and the Streamlit analytics dashboard.

5 Recommended Model for Project Implementation

groq

GPT-OSS 20B through Groq Recommended

Offers a good balance of low API cost, high inference speed and structured-output support. It can be used with separate prompts for the four evaluation agents (**Relevance**, **Accuracy**, **Hallucination Detection**, and **Completeness**), while the **Verdict Agent** calculates the weighted score and consolidated Pass / Needs Improvement / Fail reasoning in Python.

Note: GPT-OSS 120B and Gemini Flash should be tested as alternatives before finalizing the model (supporting Milestone 4's requirement to showcase evaluation across distinct AI systems). The recommendation is provisional as no project-specific benchmark has been completed.

6 Conclusion

GPT-OSS 20B through Groq is a suitable initial choice for the **AI Response Quality Evaluator Agent**. It combines fast inference, low API cost and structured-output support and integrates well with the RAG pipeline, sentence embeddings, ChromaDB and Python-based scoring.

The final model should be validated using a fixed dataset (such as seeded **TruthfulQA** and **SQuAD** benchmarks from Hugging Face) containing correct, incomplete, irrelevant and hallucinated answers. Comparing evaluation accuracy, hallucination detection, consistency, latency and cost will help determine the most suitable model for the completed system.